# ADAM LISTER | CV

adamlisteruk@proton.me • adam-lister.com

## CURRENT

**Position** Scientist I, Univeristy of Wisconsin – Madison

**Bio** Expertise in New Physics Searches, Neutrino Oscillations, Detector Physics, and Detector Calibration. Collaborator on two international neutrino experiments, NOvA (200 members) and DUNE (1600 members)

**Leadership** NOvA NuX (New Physics in Oscillations) Convener

**Projects** *BNB Neutrinos at NOvA* • Leader in understanding neutrinos from NOvA's second beamline, including flux simulation, integration into NOvA software, and developing data quality requirements.

*OSCAr Test Stand* • Developing test stand to measure never-before-measured transverse component of diffusion of ionisation electrons in LArTPCs.

## EDUCATION AND EMPLOYMENT

2024-present **Scientist I** *University of Wisconsin – Madison, USA*
Collaborator on the NOvA and DUNE experiments

2019-2024 **Postdoctoral Research Associate** *University of Wisconsin – Madison, USA*
Collaborator on the NOvA and DUNE experiments

2015-2019 **PhD Experimental Particle Physics** *Lancaster University, UK*
Collaborator on the MicroBooNE experiment
Thesis: Constraint of Systematic Uncertainties in an Electron Neutrino Search Using Muon Neutrinos at MicroBooNE

2013-2014 **MSc Particles, Strings, and Cosmology** *Durham University, UK*
Thesis: Neutrino Oscillations in Long-Baseline Experiments

2010-2013 **BSc Astrophysics (1st class)** *Aberystwyth University, UK*

## LEADERSHIP

2024-present **NOvA NuX Group Convener** (appointed) ∼30 people
Managing searches for New Physics in neutrino oscillations • Two PRL publications during tenure, with another in internal review • Seven active analyses, each targeting high-impact journals.

2025-2026 **NOvA Executive Committee** (elected)
Elected to advise the spokespeople on science and management.

2022-2024 **NOvA Detector Systematics Group Convener** (appointed) ∼30 people
Delivered simulation improvements (Geant4, neutron model, detector simulation) • Updated calibration scheme which increased speed and reduced failure rates.

2020-2022 **NOvA Production Group Convener** (appointed) ∼5 people
Delivered largest ever file production used for all recent NOvA publications • Developed computing infrastructure & monitoring tools • Developed new storage-informed production model, enabling backprocessing of ∼ 1 PB of cosmic-ray data which was used in NOvA's flagship neutrino oscillation search.

## AWARDS, OUTREACH & SERVICE

**Awards** NOvA Core Service Award (’26) • Contributor to successful UW-Madison DOE Base Grants (’20,’23,’26) • $10,000 Fermilab Neutrino Physics Center Fellowship (’18) • Lancaster University Department Prize (’19)

**Outreach** NOvA public website manager (’24-present) • NOvA social media manager (’24-present) • Fermilab LGBTQ+ group Pride Parade (’22,’24) • NOvA Experiment Facility Tour Guide (’23-’24) • Coordinated and featured in articles for APS Physics Magazine and Physics World for NOvA Sterile Neutrino Results • Fermilab Open House volunteer (’17,’19)

**Service** NOvA Paper Review Committee for neutrino interactions measurement (’26) • NOvA Analysis Review Committee for cosmic-ray muon measurement (’23-’25) • Session chair at Lepton-Photon 2025 • NOvA Workshop organiser (’24,’25) • DUNE Plot Style Committee (’22-’23)

## PRESENTATIONS

Jun 2026 **Neutrino 2026** *Poster*
Future Sterile Neutrino Searches in NOvA using the Booster Neutrino Beam

Jan 2026 **NuPhys 2026** *Contributed Talk*
NOvA’s Current and Future Sterile Neutrino Searches

Aug 2025 **Lepton-Photon 2025** *Contributed Talk*
Beyond Standard Model Neutrino Oscillation Results from NOvA

Jun 2025 **WIN 2025** *Contributed Talk*
Beyond Standard Model Neutrino Oscillation Results from NOvA

Apr 2025 **U. Chicago, U. Cincinnati, U. Mississippi** *Invited Colloquia/Seminars*
Looking for Sterile Neutrinos with the NOvA Experiment

Jun 2024 **Neutrino 2024** *Poster*
Improving NOvA’s Sterile Neutrino Search with the Booster Neutrino Beam

Dec 2023 **NuPhys 2023** *Invited Talk*
Neutrino Oscillations at NOvA

Sep 2023 **Cornell University** *Invited Seminar*
Searching For Sterile Neutrinos With The NOvA Experiment

Dec 2022 **CPAD Workshop 2022** *Contributed Talk*
Effects of Ionisation Electron Diffusion on Calibrations in LArTPCs

Jun 2022 **55th Annual Fermilab Users’ Meeting** *Invited Talk*
Probing Neutrino Oscillations at The NOvA Experiment

Jun 2020 **Neutrino 2020** *Poster*
Event Selection and Systematics for the NOvA Sterile Neutrino Search

Dec 2018 **University of Wisconsin – Madison** *Invited Seminar*
MicroBooNE and the Path to Resolving the MiniBooNE Low-Energy Excess

Mar 2018 **IOP HEPP and APP Meeting** *Contributed Talk*
Measuring Electron Diffusion in MicroBooNE

Jun 2017 **50th Annual Fermilab Users’ Meeting** *Poster*
Towards a Longitudinal Electron Diffusion Measurement in MicroBooNE

Jul 2016 **Neutrino 2016** *Poster*
Measurements of $\nu_\mu$ CC Interactions by the MicroBooNE Experiment

## SELECT PUBLICATIONS

Listed below are those publications where I have had a significant contribution. A full list of publications can be found on ORCiD, or on iNSPIRE.

2026 **NOvA Dual-Baseline Search for Active-to-Sterile Neutrino Oscillations using Neutrino- and Antineutrino-Enriched Samples**
*The NOvA Collaboration, arXiv 2609.08900*

2026 **NOvA's Current and Future Sterile Neutrino Searches**
*Adam Lister, for the NOvA Collaboration, arXiv 2602.11345*

2026 **Precision measurement of neutrino oscillation parameters with 10 years of data from the NOvA experiment**
*The NOvA Collaboration, PRL 136 011802*

2026 **Explanation of the Seasonal Variation of Cosmic Multiple Muon Events Observed with the NOvA Near Detector**
*The NOvA Collaboration, PRD 113 012001*

2025 **Dual-Baseline Search for Active-to-Sterile Neutrino Oscillations in NOvA**
*The NOvA Collaboration, PRL 134 081804*

2024 **Search for CP-Violating Neutrino Nonstandard Interactions with the NOvA Experiment**
*The NOvA Collaboration, PRL 133 201802*

2022 **Investigations on a fuzzy process: effect of diffusion on calibration and particle identification in Liquid Argon Time Projection Chambers**
*A. Lister & M. Stancari, JINST 17 P07016*

2022 **Search for an Excess of Electron Neutrino Interactions in MicroBooNE Using Multiple Final-State Topologies**
*The MicroBooNE Collaboration, PRL 128 241801*

2022 **Search for an anomalous excess of charged-current $\nu_e$ interactions without pions in the final state with the MicroBooNE experiment**
*The MicroBooNE Collaboration, PRD 105 112004*

2021 **Measurement of the longitudinal diffusion of ionization electrons in the MicroBooNE detector**
*The MicroBooNE Collaboration, JINST 16 P09025*

2020 **Measurement of differential cross sections for muon neutrino-argon charged-current interactions with protons and no pions in the final state in the MicroBooNE detector**
*The MicroBooNE Collaboration, PRD 102 112013*

2020 **Deep Underground Neutrino Experiment (DUNE), Far Detector Technical Design Report, Volume IV: Far Detector Single-phase Technology**
*The DUNE Collaboration, JINST 15 T08010*